CSCE 5300: Introduction to Big Data and Data Science

Chapter 5 (5.3): Data Learning Methods

Outline

- 5.3.1. Introduction
- 5.3.2. Algorithm of K-Means Clustering
- 5.3.3. Elbow method
- 5.3.4. Computer simulations
- 5.3.5. Advantages and limitations
- 5.3.6. Applications

5.3.1. Introduction

- ***K-Means Clustering is an unsupervised learning algorithm*** *that is used to solve the clustering problems in machine learning or data science.*
- *K-means clustering aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest mean, serving as a prototype of the cluster.*
- *Here K defines the number of pre-defined clusters that need to be created in the process, as if K=2, there will be two clusters, and for K=3, there will be three clusters, and so on.*
- *It is an iterative algorithm that divides the unlabeled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.*

5.3.1. Introduction

- *It allows us to cluster the data into different groups and a convenient way to discover the categories of groups in the unlabeled dataset on its own* ***without the need for any training****.*

- It is a centroid-based algorithm, where each cluster is associated with a centroid.
- Because each object in this example has two attributes, it is useful to consider each object corresponding to the point *(x,. y),* where x and y denote the two attributes and i = 1, 2 ... M.
- *For a given cluster of m points (m <= M), the point that corresponds to the cluster's mean is called* ***a centroid****.*

5.3.1. Introduction

- *In mathematics, a centroid refers to a point that corresponds to the center of mass for an object.*
- *The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding centroids.*
- *The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters.*
- *The value of k should be predetermined in this algorithm.*

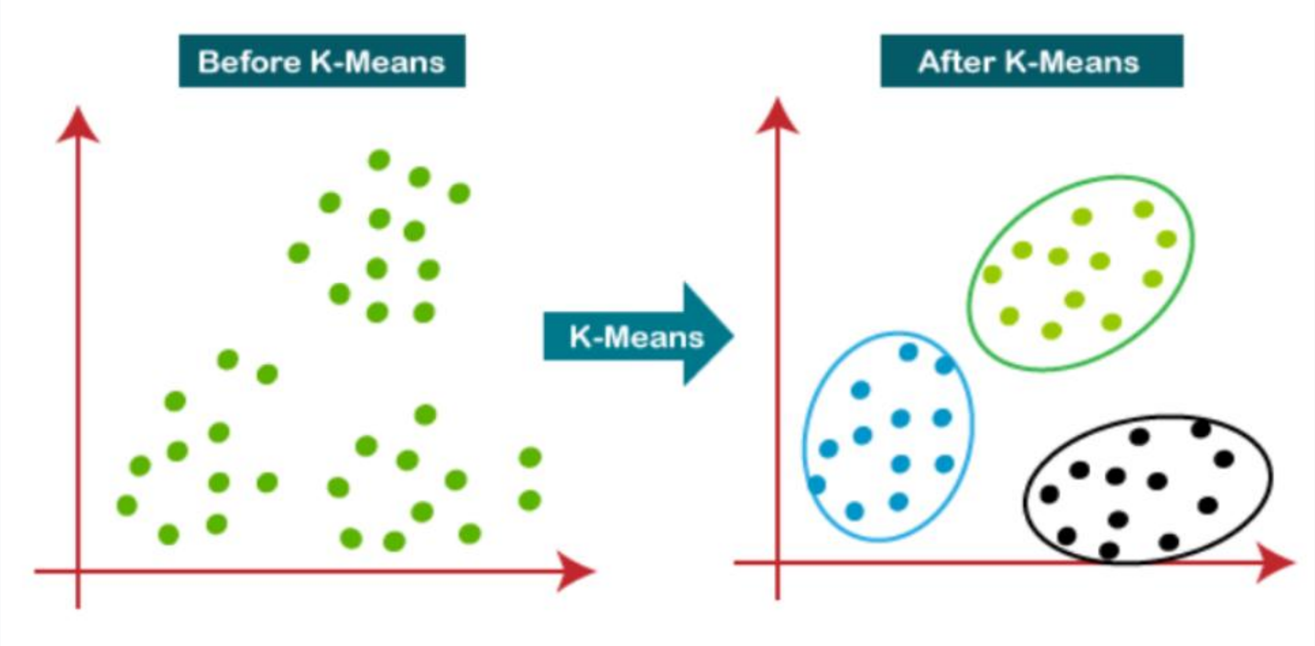
5.3.1. Introduction

- The k-means clustering algorithm mainly performs two tasks:
- *Determines the best value for K center points or centroids by an iterative process.*
- *Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster* [1], [2].
- The below diagram explains the working of the K-means Clustering Algorithm (Fig. 5.3.1):

- [1] Shai Shalev-Shwartz and Shai Ben-David, *Understanding Machine learning: From theory to Algorithms,* Cambridge University Press, New York, 2014.
- [2] Alex Smola and S.V.N. Vishwanathan, *Introduction to Machine learning,* Cambridge University Press, Cambridge, UK, 2010.

5.3.1. Introduction

- Fig. 5.3.1



5.3.1. Introduction

- To illustrate the method to find k clusters from a collection of M objects with n attributes, the two-dimensional case (n = 2) is examined.
- *It is much easier to visualize the k-means method in two dimensions.*
- ***In two dimensions, the distance between any two points in the Cartesian plane is typically expressed by using the Euclidean distance measure***
- $$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \qquad (5.3.1)$$
- ***In two dimensions, the centroid of them points in a k-means cluster is calculated as follows***
- $$(x_c, y_c) = \left(\frac{\sum_{i=1}^{m} x_i}{m}, \frac{\sum_{i=1}^{m} y_i}{m} \right) \qquad (5.3.2)$$

5.3.1. Introduction

- *Given a set of observations (x_1, x_2, …, x_n), where each observation is a d-dimensional real vector, k-means clustering aims to partition the n observations into k groups G = {G_1, G_2, …, G_k} so as to minimize the within-cluster sum of squares defined as follows:*

$$argmin \sum_{i=1}^{k} \sum_{x \in S_i} \| x - \mu_i \|^2 \qquad (5.3.3)$$

- The later formula shows the objective function that is minimized in order to find the optimal prototypes in k-means clustering.

5.3.2. Algorithm

- *The intuition of the formula is that we would like to find groups that are different with each another and each member of each group should be similar with the other members of each cluster.*

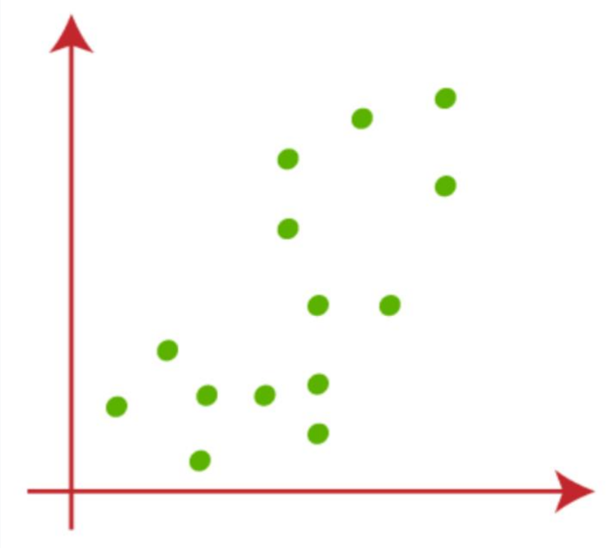
- **Algorithm of K-Means Clustering**
- ***Step-1:*** *Select the number K to decide the number of clusters.*
- ***Step-2:*** *Select random K points or centroids. (It can be other from the input dataset).*

5.3.2. Algorithm

- ***Step-3:*** *Assign each data point to their closest centroid, which will form the predefined K clusters.*
- ***Step-4:*** *Calculate the variance and place a new centroid of each cluster.*
- ***Step-5:*** *Repeat the third step, which means reassign each data point to the new closest centroid of each cluster.*
- ***Step-6:*** *If any reassignment occurs, then go to step-4 else go to FINISH.*
- ***Step-7****: The model is ready.*

5.3.2. Algorithm

- Let's understand the above steps by considering the visual plots:
- Suppose we have two variables M1 and M2.
- The x-y axis scatter plot of these two variables is given below (Fig. 5.3.2):



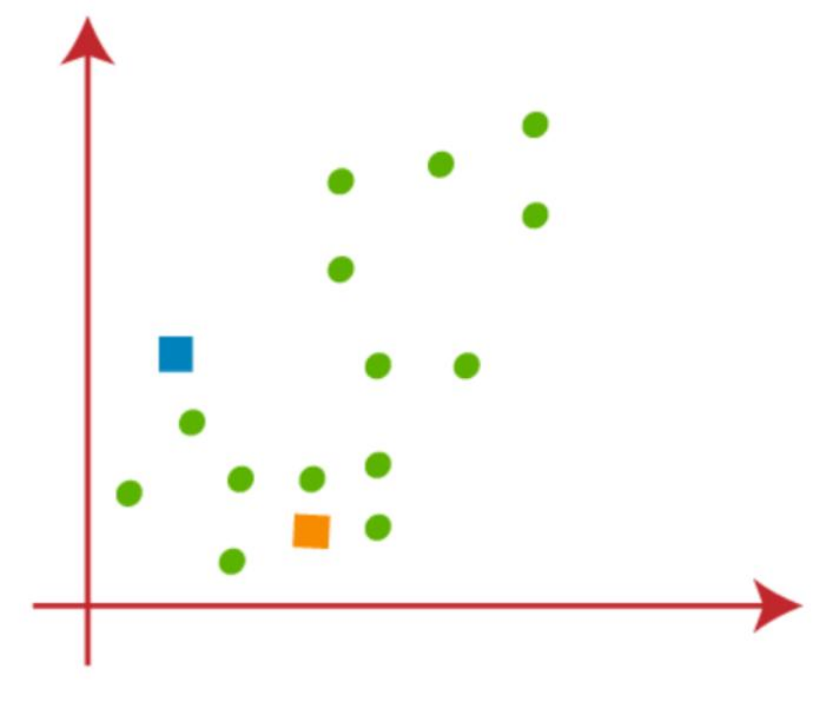
- Fig. 5.3.2

5.3.2. Algorithm

- *Let's take number k of clusters, i.e., K=2, to identify the dataset and to put them into different clusters.*
- *It means here we will try to group these datasets into two different clusters.*

- *We need to choose some random k points or centroid to form the cluster.*
- *These points can be either the points from the dataset or any other point.*
- *So, here we are selecting the below two points as k points, which are not the part of our dataset.*
- Consider the below image (Fig. 5.3.3):

5.3.2. Algorithm



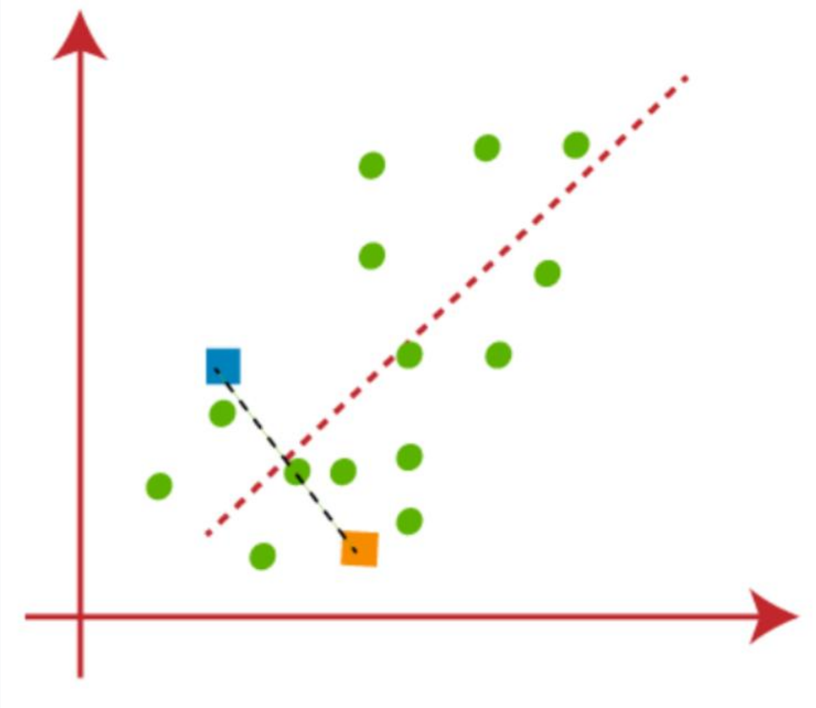
- Fig. 5.3.3

5.3.2. Algorithm

- *Now we will assign each data point of the scatter plot to its closest K-point or centroid.*
- We will compute it by applying some mathematics that we have studied to calculate the distance between two points.
- *So, we will draw a median between both the centroids.*
- Consider the below image (Fig. 5.3.4):

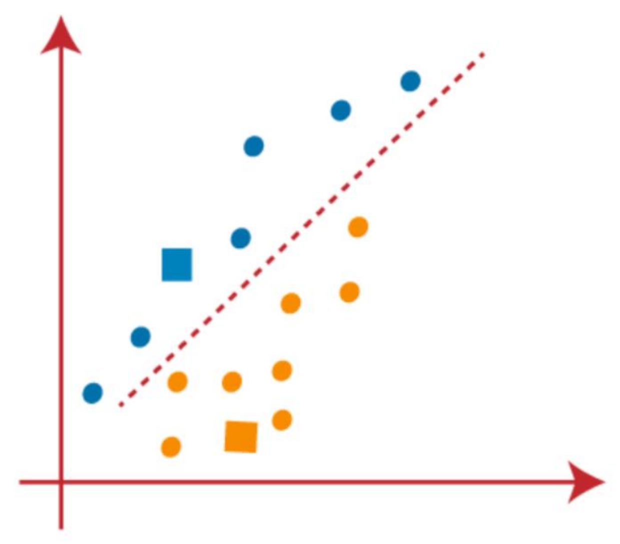
5.3.2. Algorithm

- Fig. 5.3.4



5.3.2. Algorithm

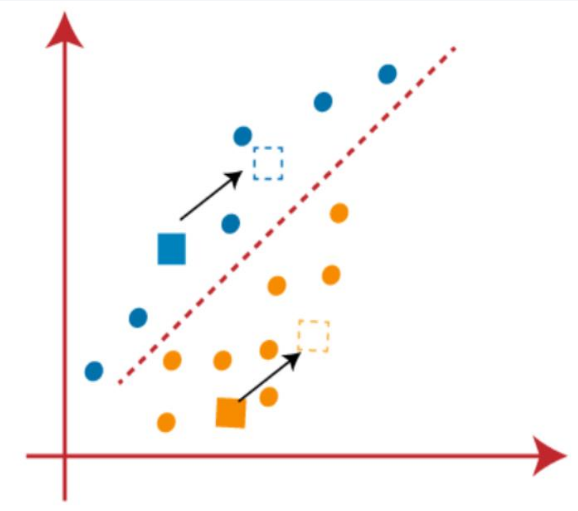
- *From the above image, it is clear that points left side of the line are near to the K1 or blue centroid, and points to the right of the line are close to the yellow centroid.*
- *Let's color them as blue and yellow for clear visualization* (Fig. 5.3.5).



- Fig. 5.3.5

5.3.2. Algorithm

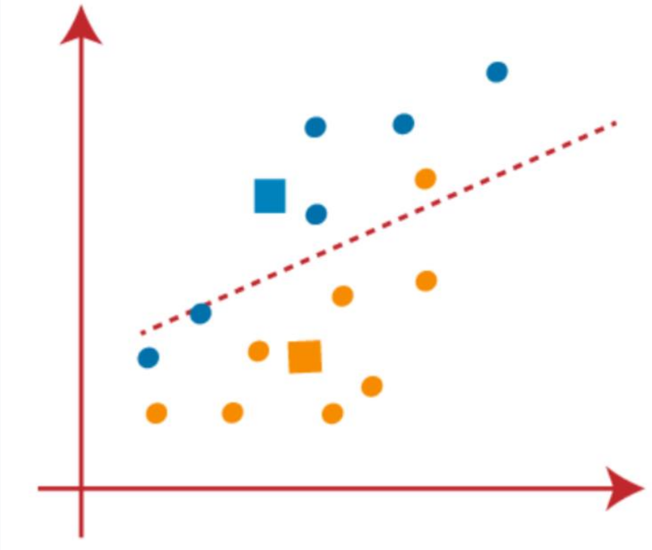
- *As we need to find the closest cluster, so we will repeat the process by choosing* ***a new centroid****.*
- *To choose the new centroids, we will compute the center of gravity of these centroids, and will find new centroids as below* (Fig. 5.3.6):
- Fig. 5.3.6



5.3.2. Algorithm

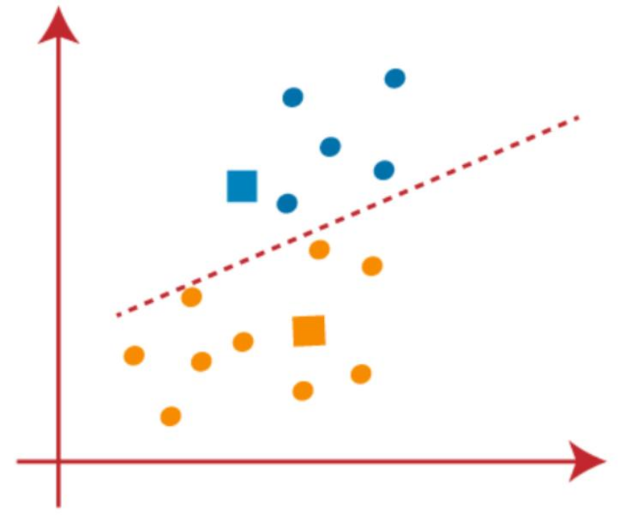
- *Next, we will reassign each data point to the new centroid.*
- *For this, we will repeat the same process of finding a median line.*
- *The median will be like below image* (Fig. 5.3.7):

- Fig. 5.3.7



5.3.2. Algorithm

- *From the above image, we can see, one yellow point is on the left side of the line, and two blue points are right to the line.*
- *So, these three points will be assigned to new centroids* (Fig. 5.3.8).



- Fig. 5.3.8

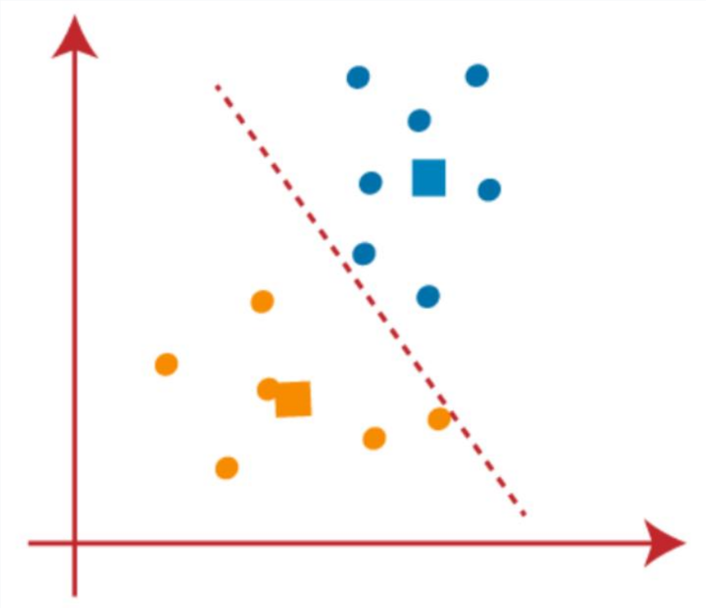
5.3.2. Algorithm

- *As reassignment has taken place, so we will again go to the step-4, which is finding new centroids or K-points.*
- *We will repeat the process by finding the center of gravity of centroids, so the new centroids will be as shown in the below image* (Fig. 5.3.9):

- Fig. 5.3.9

5.3.2. Algorithm

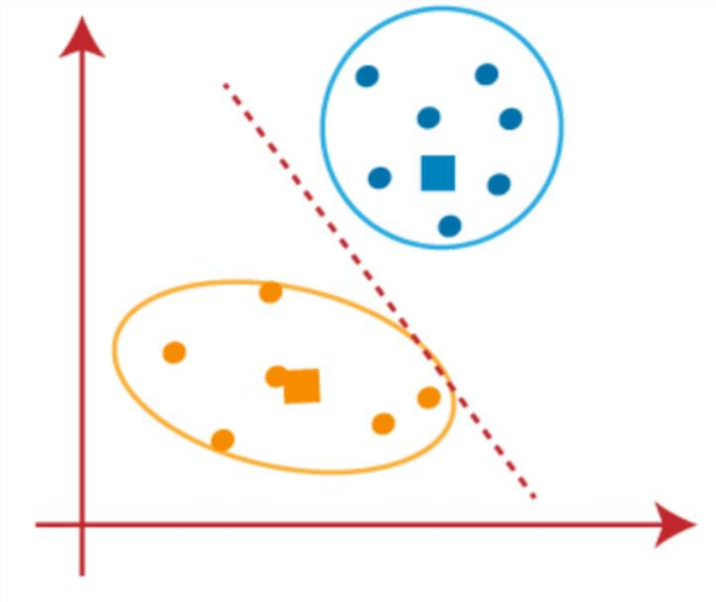
- *As we got the new centroids so again will draw the median line and reassign the data points.*
- *So, the image will be* (Fig. 5.3.10):



- Fig. 5.3.10

5.3.2. Algorithm

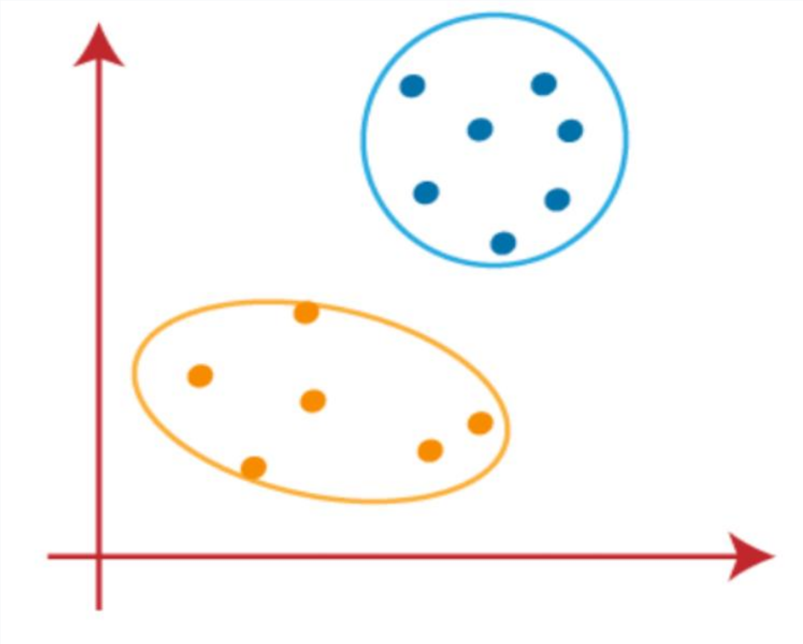
- *We can see in the above image; there are no dissimilar data points on either side of the line, which means our model is formed.*
- Consider the below image (Fig. 5.3.11):



- Fig. 5.3.11

5.3.2. Algorithm

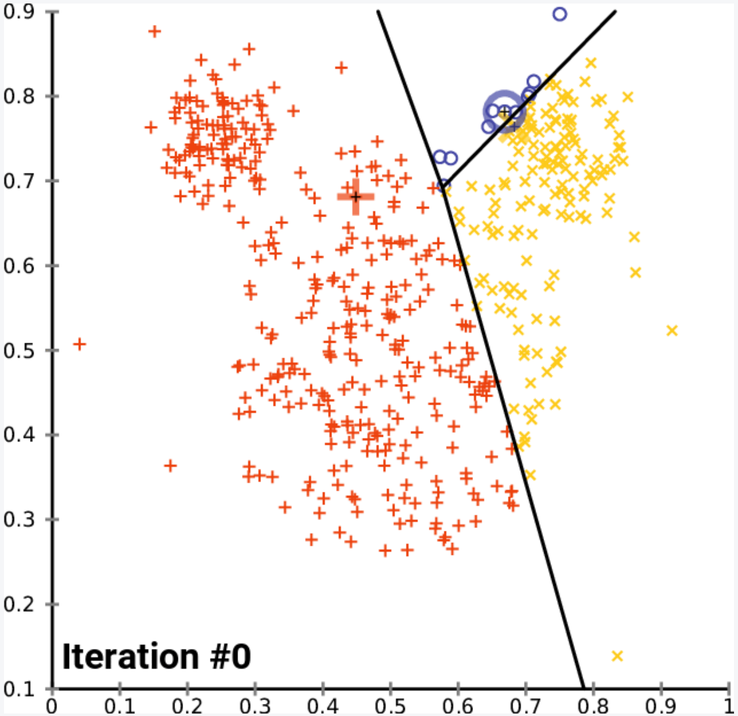
- *As our model is ready, so we can now remove the assumed centroids, and the two final clusters will be as shown in the below image* (Fig. 5.3.12):



- Fig. 5.3.12

5.3.2. Algorithm

- Fig. 5.3.13



5.3.3. Elbow Method

- **How to choose the value of "K number of clusters" in K-means Clustering?**
- *The performance of the K-means clustering algorithm depends upon highly efficient clusters that it forms.*
- *But choosing the optimal number of clusters is a big task.*
- *There are some different ways to find the optimal number of clusters, but here we are discussing the most appropriate method to find the number of clusters or value of K.*
- *The method is given below:*

5.3.3. Elbow Method

- The Elbow method is one of the most popular ways to find the optimal number of clusters.
- This method uses the concept of WCSS value.
- ***WCSS*** *stands for* ***Within Cluster Sum of Squares****, which defines the total variations within a cluster.*
- *The formula to calculate the value of WCSS (for 3 clusters) is given below:*
- $$\mathrm{WCSS} = \sum_{P_i \text{ in Cluster1}} \mathrm{distance}(P_i\ C_1)^2 + \sum_{P_i \text{ in Cluster2}} \mathrm{distance}(P_i\ C_2)^2 + \sum_{P_i \text{ in CLuster3}} \mathrm{distance}(P_i\ C_3)^2 \qquad (5.3.4)$$
-

5.3.3. Elbow Method

- In the above formula of WCSS,
- $\sum_{P_i \text{ in Cluster1}}$ *distance*$(P_i\ C_1)^2$*: It is the sum of the square of the distances between each data point and its centroid within a cluster1 and the same for the other two terms.*
- *To measure the distance between data points and centroid, we can use any method such as Euclidean distance or Manhattan distance.*

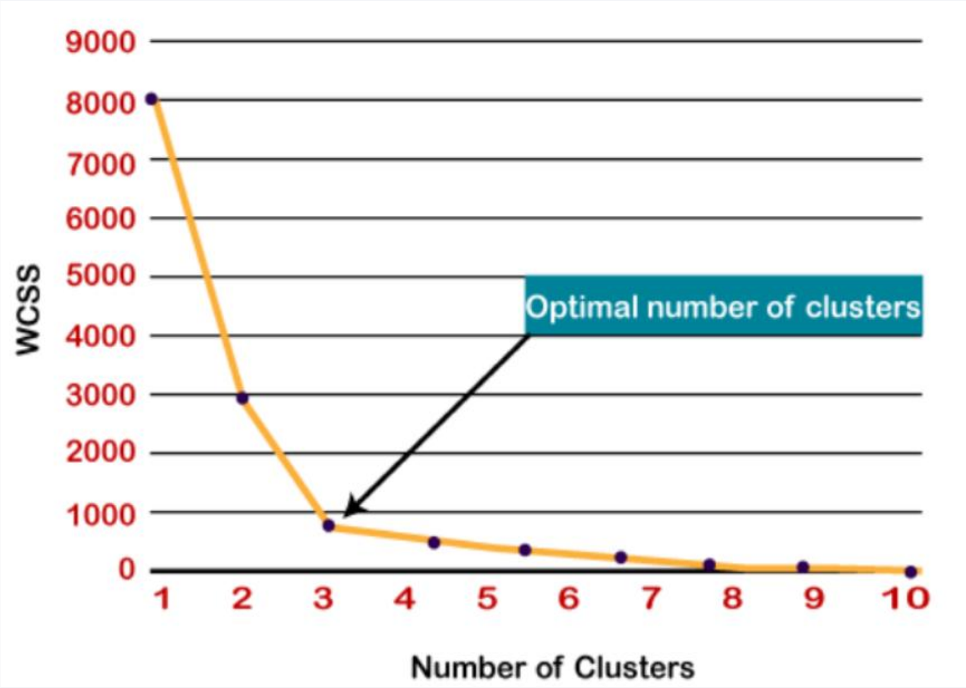
- *In n-dimensional case, (5.3.4) will be given by*
- $$WSS = \sum_{i=1}^{M} d(p_i, q^{(i)})^2 = \sum_{i=1}^{M}\sum_{j=1}^{n}\left(p_{ij} - q_j^{(i)}\right)^2 \qquad (5.3.5)$$

5.3.3. Elbow Method

- To find the optimal value of clusters, the elbow method follows the below steps:
- *It executes the K-means clustering on a given dataset for different K values (ranges from 1-10).*
- *For each value of K, calculates the WCSS value.*
- *Plots a curve between calculated WCSS values and the number of clusters K.*
- *The sharp point of bend or a point of the plot looks like an arm, then that point is considered as the best value of K.*
- *Since the graph shows the sharp bend, which looks like an elbow, hence it is known as the elbow method.*
- The graph for the elbow method looks like the below image (Fig. 5.3.14):

5.3.3. Elbow Method

- Fig. 5.3.14



5.3.4. Computer simulations

- The python code and output results can be presented as follows:

- #importing the required Python libraries:
- import numpy as np
- from numpy import vstack,array
- from numpy.random import rand
- from scipy.cluster.vq import whiten, kmeans, vq
- from pylab import plot,show

- #Random data generation:
- data = vstack((rand(200,2) + array([.5,.5]),rand(150,2)))

- #Normalizing the data:
- data = whiten(data)

5.3.4. Computer simulations

- # computing K-Means with K = 3 (3 clusters)
- centroids, mean_value = kmeans(data, 3)
- print("Code book :\n", centroids, "\n")
- print("Mean of Euclidean distances :", mean_value.round(4))

- # mapping the centroids
- clusters, _ = vq(data, centroids)
- print("Cluster index :", clusters, "\n")

5.3.4. Computer simulations

- #Plotting using numpy's logical indexing
- plot(data[clusters==0,0],data[clusters==0,1],'ob',
- data[clusters==1,0],data[clusters==1,1],'or',
- data[clusters==2,0],data[clusters==2,1],'og')

- plot(centroids[:,0],centroids[:,1],'sg',markersize=8)

- show()

5.3.4. Computer simulations

- Code book :
- [[3.45722219 2.644075]
- [1.59695324 1.11172277]
- [1.95075508 2.77699435]]

- Mean of Euclidean distances : 0.802 (Fig. 5.29) (Fig. 5.43)
- Cluster index : [2 0 0 0 0 2 1 2 0 2 0 2 0 2 1 2 1 0 0 2 2 2 1 1 0 0 2 1 2 0 0 0 0 2 0 2 0
- 2 0 2 2 0 1 0 2 2 2 0 0 2 0 1 2 2 2 0 0 1 2 2 0 0 0 2 0 0 2 2 2 2 0 0 0 0
- 0 0 2 0 2 2 2 2 2 0 0 0 2 0 0 2 0 2 0 0 0 2 2 0 1 2 2 0 0 0 0 2 2 0 1 0 0
- 0 2 0 2 0 0 0 0 2 2 2 0 1 1 0 0 2 0 0 0 0 0 0 2 0 2 2 0 2 0 2 0 2 0 2 2 0
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- 0 2 0 2 2 2 1 0 2 2 0 0 0 0 0 1 1 1 1 1 0 2 2 1 1 1 1 2 2 1 2 1 1 1 1 1 2
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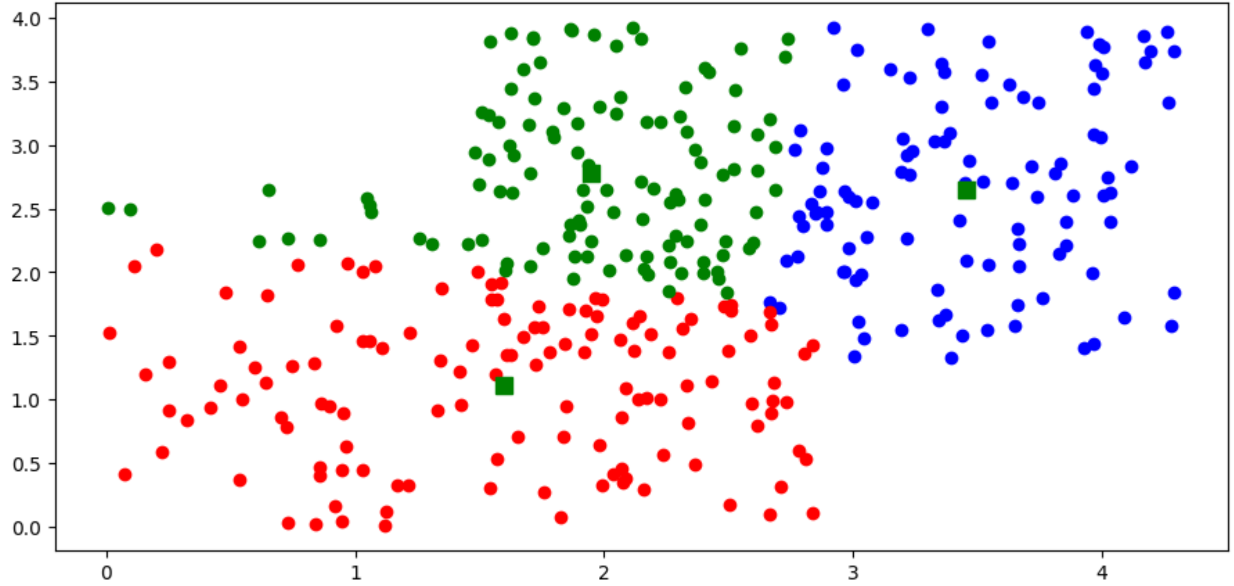
- 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 2 1 1 2 1
- 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 2 1 2 1 1 1 2 2 2 2 2 1 1 1 1 1 0 1 1
- 1 1 1 1 2 1 1 1 1 0 1 1 2 2 1 2 1]

5.3.4. Computer simulations

- Fig. 5.3.15



5.3.5. Advantages and Limitations

- **Pros:**
  1. Simple: It is easy to implement k-means and identify unknown groups of data from complex data sets. The results are presented in an easy and simple manner.
  2. Flexible: K-means algorithm can easily adjust to the changes. If there are any problems, adjusting the cluster segment will allow changes to easily occur on the algorithm.
- *3. Suitable in a large dataset: K-means is suitable for a large number of datasets and it’s computed much faster than the smaller dataset. It can also produce higher clusters.*
  4. Efficient: The algorithm used is good at segmenting the large data set. Its efficiency depends on the shape of the clusters. K-means works well in hyper-spherical clusters.

5.3.5. Advantages and Limitations

- *5. Time complexity: K-means segmentation is linear in the number of data objects thus increasing execution time. It doesn't take more time in classifying similar characteristics in data like hierarchical algorithms.*
- *6. Tight clusters: Compared to hierarchical algorithms, k-means produce tighter clusters especially with globular clusters.*
  7. Easy to interpret: The results are easy to interpret. It generates cluster descriptions in a form minimized to ease understanding of the data.
- *8. Computation cost: Compared to using other clustering methods, a k-means clustering technique is fast and efficient in terms of its computational cost O(K*n*d).*
- *9. Accuracy: K-means analysis improves clustering accuracy and ensures information about a particular problem domain is available. Modification of the k-means algorithm based on this information improves the accuracy of the clusters.*

5.3.5. Advantages and Limitations

- **Cons:**
  1. NoNo-optimal set of clusters: K-means doesn’t allow the development of an optimal set of clusters and for effective results, you should decide on the clusters before.
- *2. Lacks consistency: K-means clustering gives varying results on different runs of an algorithm. A random choice of cluster patterns yields different clustering results resulting in inconsistency.*
  3. Order of values: The way in which data is ordered in building the algorithm affects the final results of the data set.

5.3.5. Advantages and Limitations

- *5. Sensitivity to scale: Changing or rescaling the dataset either through normalization or standardization will completely change the final results.*
- *6. Crash computer: When dealing with a large dataset, conducting a dendrogram technique will crash the computer due to a lot of computational load and Ram limits.*
- *7. Handle numerical data: K-means algorithm can be performed in numerical data only.*
- *8 Specify K-values: For K-means clustering to be effective, you have to specify the number of clusters (K) at the beginning of the algorithm.*
- *9. Prediction issues: It is difficult to predict the k-values or the number of clusters. It is also difficult to compare the quality of the produced clusters.*

5.3.6. Applications

- Clustering is applied in the following areas:

- ***Data summarization and compression:*** *Clustering is widely used in the areas where we require data summarization, compression and reduction as well.*
- *The examples are image processing and vector quantization.*

- ***Collaborative systems and customer segmentation:*** *Since clustering can be used to find similar products or same kind of users, it can be used in the area of collaborative systems and customer segmentation.*

5.3.6. Applications

- ***Serve as a key intermediate step for other data mining tasks:*** *Cluster analysis can generate a compact summary of data for classification, testing, hypothesis generation; hence, it serves as a key intermediate step for other data mining tasks also.*

- ***Trend detection in dynamic data:*** *Clustering can also be used for trend detection in dynamic data by making various clusters of similar trends.*

- ***Social network analysis:*** *Clustering can be used in social network analysis.*
- *The examples are generating sequences in images, videos or audios.*

- ***Biological data analysis:*** *Clustering can also be used to make clusters of images, videos hence it can successfully be used in biological data analysis.*